*Completed using the Barron's, College Board Required Knowledge, and the Notes.

The Math Section

(Every Single Calculation or Formula from AP Bio Curriculum)

*Remember, all formulas will be provided but you should know how to use them

Anywhere there is a red star indicates a practice section. ☆

Chi Square Test

- The Chi Square Test weighs actual results against expected results (as proposed by a null hypothesis) to determine whether an outside force or factor is affecting outcomes or not.

- **Null Hypothesis—** proposes that an outside factor has no effect on outcomes

- **Alternate Hypothesis—** proposes that an outside factor has an effect on outcomes

A Chi Square test relies on a null hypothesis.
The **ONLY** two outcomes of a chi square test are NOT proving or disproving the null hypothesis. Instead, they are:

1. Fail to reject (Chi Square does not exceed the critical value)

2. Reject (Chi Square exceeds critical value)

Important for FRQ answers

Formula:

$$\chi^2 = \sum \frac{(\text{observed}-\text{expected})^2}{\text{expected}}$$

Chi Square Symbol

Sum of (in green) for each outcome

observed: what was recorded,
expected: what results null hypothesis suggested

If χ^2 exceeds critical value, null hypothesis is rejected.

Degrees of Freedom (df) = possible outcomes − 1

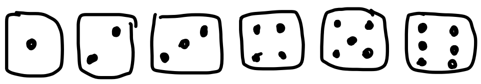
Ex: 2 outcomes =
df = 2−1
df = 1

p-val / DF	0.1	0.05	0.01	0.005
Critical values for χ^2 test				
1	2.706	3.84	6.64	7.88
2	4.605	5.99	9.21	10.60
3	6.251	7.82	11.35	12.94
4	7.779	9.49	13.28	14.86
5	9.236	11.07	15.09	16.75
6	10.65	12.59	16.81	18.55
7	12.02	14.07	18.48	20.28
8	13.36	15.51	20.09	21.96
9	14.68	16.92	21.67	23.59
10	15.99	18.31	23.21	25.19

Always use 0.05

☆

Example: You roll a dice **72** times.



18 3 14 19 10 8

Null Hypothesis: ______

Example: There is no outside factor affecting the dice, so the outcomes should be around 12 rolls per dice side.

p-val / DF	0.1	0.05	0.01	0.005
Critical values for χ^2 test				
1	2.706	3.84	6.64	7.88
2	4.605	5.99	9.21	10.60
3	6.251	7.82	11.35	12.94
4	7.779	9.49	13.28	14.86
5	9.236	11.07	15.09	16.75
6	10.65	12.59	16.81	18.55
7	12.02	14.07	18.48	20.28
8	13.36	15.51	20.09	21.96
9	14.68	16.92	21.67	23.59
10	15.99	18.31	23.21	25.19

Using a p-val of 0.05, is null hypothesis able to be rejected?

Calculation / Answer:

Answer:

$$\chi^2 = \frac{(18-12)^2}{12} + \frac{(3-12)^2}{12} + \frac{(14-12)^2}{12} + \frac{(19-12)^2}{12} + \frac{(10-12)^2}{12} + \frac{(8-12)^2}{12} =$$

$$\frac{36}{12} + \frac{81}{12} + \frac{4}{12} + \frac{49}{12} + \frac{4}{12} + \frac{16}{12} = \boxed{\chi^2 = 15.8\overline{3}}$$

df = 5
critical value = 11.07

15.83 exceeds 11.07, so null hypothesis is rejected.

Descriptive Statistics (Mean, Median)

Mean and Median are used to simply describe a data set.

Median—midpoint (number in the middle) of a data set when arranged in numerically ascending order

Mean—statistical average of all numbers in a data set

Formula = (Mean)

$$\bar{X} = \frac{\text{sum of all data points}}{\text{total number of data points}}$$

↑
Mean

Formula = (Median)

1. Rearrange numbers in ascending order
2. Find number in middle if using an odd amount of data points
3. find mean of two middlemost numbers in an even amount of data points

Example = Find **mean** and **median** of {34, 52, 89, 12, 33, 90, 22, 76, 10, 104}

Calculate/Solve:

Answer:

Mean $= \frac{34+52+89+12+33+90+22+76+10+104}{10} = \boxed{\bar{X} = 52.2}$

Median: $\{10, 12, 22, 33, 34, 52, 76, 89, 90, 104\}$, $\frac{34+52}{2} = 43$

1 2 3 4 4 3 2 1

$\boxed{\text{Median} = 43}$

Standard Deviation and Standard Error of the Mean

The Standard Deviation of the Mean determines how far each data point is from the mean of the data set

The Standard Error of the Mean is a measure of how spread out the numbers in a data set are

Formula =

current data point

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

(standard deviation of the mean)

mean

sum of all $(x_i - \bar{x})^2$

how many data points

Formula =

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

(standard error of the mean)

s ← standard deviation of the mean

$\sqrt{n}$ ← total num of data points

Example:

Calculate the standard deviation of the mean (s) and the standard error of the mean ($SE_{\bar{x}}$)
{3, 3.6, 3.2, 4, 2.1, 2.3, 2.5}

Calculate + Solve:

Answer:

$$\text{Mean} = \frac{3+3.6+3.2+4+2.1+2.3+2.5}{7} = 2.96$$

$$s = \sqrt{\frac{(3-2.96)^2+(3.6-2.96)^2+(3.2-2.96)^2+(4-2.96)^2+(2.1-2.96)^2+(2.3-2.96)^2+(2.5-2.96)^2}{7-1}}$$

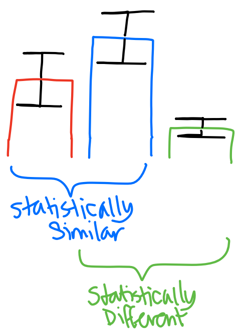
$$\boxed{s = 0.7}$$

$$SE_{\bar{x}} = \frac{0.7}{\sqrt{7}}$$

$$\boxed{SE_{\bar{x}} = 0.26}$$

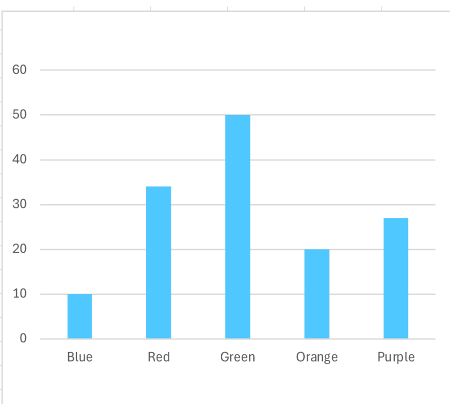
Error Bars in Bar and Line Graphs are Added to Indicate Overlap.
If two error bars overlap, then two means can be considered statistically similar. However, if they do not, they can be considered statistically different.

Error bars are calculated with +/- 2 $SE_{\bar{x}}$



Example = Add Error Bars.

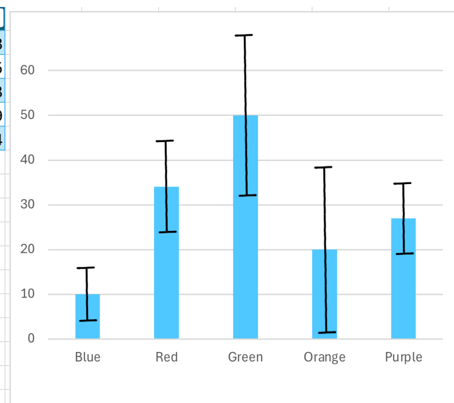
Color	Mean	$SE_{\bar{x}}$
Blue	10	3
Red	34	5
Green	50	8
Orange	20	9
Purple	27	4



Answer=

Color	Mean	$SE_{\bar{x}}$
Blue	10	3
Red	34	5
Green	50	8
Orange	20	9
Purple	27	4

Remember
+/- $\underline{\underline{2}}$ $SE_{\bar{x}}$



Water Potential/Solute Potential

Water Potential is a measure of potential energy in water.
Solute potential is a measure of how much free energy is stored in solute and is thus not available for water (always negative or zero)
measured in bars

Formula: (Water Potential)

$$\Psi = \Psi_s + \Psi_p$$

(water potential) (solute potential) (pressure potential)

Formula: (Solute potential)

$$\Psi_s = -iCRT$$

i: (ionization constant) num of particles solute breaks into
C: (molar concentration/ molarity)
R: (0.0831)
T: (temp in kelvin / Celsius + 273°)

Example: Assuming a glucose-water mixture in an open beaker in a room at 22°C with a 0.556 M molar concentration, calculate water potential.

Calculate/Solve:

Answer:

$\Psi_p = 0$ (no pressure potential, it's an open beaker.

$\Psi_s = -1(0.556)(0.0831)(22+273)$

1: glucose splits into 1 molecule

$\Psi_s = -1.36$

$\Psi = \Psi_s + \Psi_p$

$\boxed{\Psi = -1.36 \text{ bars}}$

Surface-Area-to-Volume-Ratios

The bigger an object, the smaller its surface area to volume ratio.
Cells are small because they want a high surface area to volume ratio.

Formula: (Surface Area, Rectangular Prism)

$$SA = 2lw + 2wh + 2lh$$

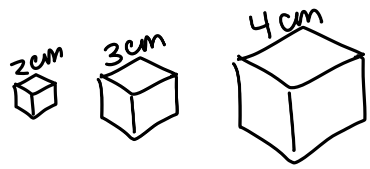
l = length
w = width
h = height

Formula: (Volume, Rectangular Prism)

$V = lwh$

l = length
w = width
h = height

Example: Calculate the surface-area-to-volume ratios of the three cubes below



Calculate/Solve:

Answer:

$SA = 2(2)(2) + 2(2)(2) + 2(2)(2)$
$SA = 24\ cm^2$
$V = (2)(2)(2)$
$V = 8\ cm^3$

$\boxed{\text{2 cm cube: SA:V } 3:1}$

$SA = 2(3)(3) + 2(3)(3) + 2(3)(3)$
$SA = 54\ cm^2$
$V = (3)(3)(3)$
$V = 27\ cm^3$

$\boxed{\text{3cm cube SA:V ratio: } 2:1}$

$SA = 2(4)(4) + 2(4)(4) + 2(4)(4)$
$SA = 96\ cm^2$
$V = (4)(4)(4)$
$V = 64\ cm^3$

$\boxed{\text{4 cm cube SA:V} = 3:2 \text{ or } 1.5:1}$

Hardy-Weinberg Equilibrium

When a population is in Hardy-Weinberg Equilibrium (no mutations, no gene flow, no mutations, large population, and random mating), these two equations should always be true.

Formula:

$p + q = 1$

p: dominant allele frequency (num of dom alleles/total alleles in population/gene pool)

q: recessive allele frequency

Formula:

$$\underbrace{p^2}_{\text{homozygous dominant frequency}} + \underbrace{2pq}_{\text{heterozygous frequency}} + \underbrace{q^2}_{\text{homozygous recessive frequency}} = 1$$

Example: Using this population, calculate the **p** (dominant allele frequency) of the population assuming it is in Hardy-Weinberg Equilibrium. A red flower displays the recessive phenotype.

Calculate/Solve:

Answer:

$q^2 = 0.25$

$q = 0.5$

$q + p = 1$

$\boxed{p = 0.5}$

Population Growth Problems

dN/dt is equal to the net change in individuals over a single generation

The **logistic growth** formula measures a population's growth, factoring in its approach to the carrying capacity (K).
The **exponential growth** formula measures a population's growth, assuming it has unlimited resources and no limiting factors.

Formula:

$$\frac{dN}{dt} = B - D$$

B: amount of births in one generation

D: amount of deaths in one generation

Formula: (Exponential Growth)

$$\frac{dN}{dt} = r_{max}(N)$$

r_{max}: maximum rate of population growth (biotic potential)

N: current pop. num

Formula: (Logistic Growth)

$$\frac{dN}{dt} = r_{max}(N)\left(\frac{K-N}{K}\right)$$

carrying capacity

Example: Assuming a population of 624 deer is exhibiting **logistic growth,** calculate the carrying capacity if the population's biotic potential is 0.44 and in the last generation 105 deer have been born and 33 deer have died.

Calculate/Solve:

Answer: $\frac{dN}{dt} = 105-33$

XXX Cannot Use Exponential Growth Formula Because "exhibiting **logistic growth,**"

$\frac{dN}{dt} = 72$

Logistic Growth Formula

$72 = 0.44(624)\left(\frac{K-624}{624}\right)$

$\boxed{K = 788 \text{ deer}}$

Bar Graph and Line Graph Creation

Bar Graphs should be used for separate "categories" and line graphs should be used for cohesive data points, most especially with time.

Example: Create a bar graph and a line graph from the following table.

Year	Apples (millions of tons)	Oranges (millions of tons)
2009	20	14
2010	22	16
2011	21	18
2012	19	17
2013	23	18

Draw:

Answer: